

Sarang Shigwan

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Summary

Software Engineer with experience in SQL, backend development, cloud deployment, and data-driven applications. Skilled in database design, REST APIs, PostgreSQL, Python, and AWS. Built projects involving data modeling, analytics workflows, and AI-assisted systems. Strong problem-solving skills with a passion for databases, data systems, and scalable software engineering.

Education

- **B.E. ENTC** at Marathwada Mitra Mandal's College of Engineering, Pune
Nov 2022 – May 2026 | **CGPA: 9.24 / 10**
- **H.S.C.** at Modern College of Arts, Science & Commerce, Pune
2020 – 2022 | **74.17%**
- **S.S.C.** at St. Francis De Sales School, Pune
2007 – 2020 | **89.80%**

Experience

Backend Intern(AWS Cloud) - EvolvingX Pvt. Ltd.

Oct 2024 – June 2025

- Developed a full-stack web application (React + Django) with REST APIs supporting dynamic user workflows
- Deployed production system on AWS EC2 with HTTPS and database integration
- Improved backend efficiency by optimizing API handling and middleware design
- Ensured secure communication using token-based authentication.

Projects

LexVault (Smart Legal Document Manager)

Github Repo: [Lexvault](#)

- Built a document intelligence system with version tracking and audit history
- Designed PostgreSQL schema for efficient retrieval of document versions
- Developed REST APIs + React dashboard
- Designed foundation for AI-based document search using embeddings.

Skills : FastAPI, PostgreSQL, SQL, REST API Design, Database Schema Design, Version Control Systems, Data Modeling.

TraceLens

Github Repo: [Tracelens](#) | **Design Report :** [Report](#)

- Built an AI-powered root cause analysis system using a RAG pipeline with vector embeddings to analyze distributed system failures
- Designed a FastAPI-based microservice deployed on AWS Lambda for scalable, stateless inference
- Implemented vector similarity search with cosine scoring to identify failure patterns with confidence scoring
- Enabled automated debugging workflows, reducing manual log analysis effort.

Skills : Sentence-Transformers (all-MiniLM-L6-v2), RAG, NumPy, Pydantic, Vector Similarity Search, Cosine Similarity, NLP Embeddings

HFT Latency Monitoring Platform

Github Repo: [LatencyMonitor](#) | **Design Report :** [Report](#)

- Built a sub-millisecond UDP ingestion pipeline with P50/P95/P99 latency tracking on Kubernetes.
- Implemented Prometheus histograms and Grafana dashboards for real-time performance observability.
- Containerized a multi-service Go architecture using Docker and Kubernetes, debugging runtime issues for reliable deployments.

Skills : Low-latency systems, Go concurrency, Docker, Kubernetes, Prometheus/Grafana, Linux debugging, UDP pipelines.

Skills

- **Fundamentals :** Operating Systems, DBMS, Networking, OOP, DSA, SDLC, system design
- **Technical Skills:** SQL, PostgreSQL, Database Design, Joins, Indexes, Views, Stored Procedures, Python, Pandas, NumPy, FastAPI, Django, REST APIs, AWS EC2, Lambda, S3, Git, Docker, CI/CD
- **Database & Data Skills :** Query Optimization, Data Modeling, Schema Design, Data Processing, API Integration
- **AI/Data Analytics Exposure :** NLP Embeddings, RAG Pipelines, Vector Similarity Search, AI-assisted Workflows

Achievements

- **AIR 2** SAENIS Efficycle 2024
- **Winner** at Hackathons like Udaan 2023 and Dextirity 2024
- **Presented research papers at conferences like ICEI and GConSoft**